Band-1 Hadamard Order 9840 – Classical Paley-I (q=9839); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9840 Paley-I PASS; Paley-II FAIL/SKIP (q=4919 prime but ===3 mod 4; Paley-II needs ===1 mod 4); Kronecker FAIL/SKIP (4*2460; 2460===0 mod 4 admissible but NOT in ONBOX_SEEDS); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-22

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-22_Band1_Hadamard_9812_PaleyI`; `2026-09-22_Band1_Hadamard_9804_PaleyI`; `2026-09-22_Band1_Hadamard_9792_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9840_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9840 only (verified max|HHˆT-nI|=0), via classical Paley-I q=9839 (prime ===3 mod 4). 9840 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9840: Paley-I **PASS** ($q = 9839$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 4919$ prime but $\equiv 3 \pmod 4$; Paley-II needs $\equiv 1 \pmod 4$); Kronecker **FAIL/SKIP** (note $9840 = 4 \cdot 2460$; $2460 \equiv 0 \pmod 4$ admissible but NOT in ONBOX_SEEDS). **9840** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9812+9804+9792+9788+9768+9744+9740+9720+9680+9644+9632+9624+9620+9588+9552+9548-** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **9840**.

- H_{9840} = Paley-I ($q = 9839$) – classical SPK

- Paley-II for 9840: **FAIL/SKIP** ($q = 4919$ prime but $\equiv 3 \pmod 4$; Paley-II needs $\equiv 1 \pmod 4$)
- Kronecker for 9840: **FAIL/SKIP** ($4 \cdot 2460$; $2460 \equiv 0 \pmod 4$ admissible but NOT in ONBOX_SEEDS)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9840.py
```

Results. overall_pass=True; units 164/164; row blocks 77/77 (block size 128); wall 761.213s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9840 = q + 1$ with $q = 9839$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 4919$ prime but $\equiv 3 \pmod 4$; Paley-II needs $\equiv 1 \pmod 4$ — FAIL/SKIP.
- Kronecker: $9840 = 4 \cdot 2460$; $2460 \equiv 0 \pmod 4$ admissible but NOT in ONBOX_SEEDS — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9840 Paley-I	Paley-I q=9839	0	752.6591
H_9840 Paley-II	FAIL/SKIP (q=4919 prime but ===3 mod 4; Paley-II needs ===1 mod 4)	–	–
H_9840 Kronecker	FAIL/SKIP (4*2460; 2460===0 mod 4 admissible but NOT in ONBOX_SEEDS)	–	–

SPK closure note

Seeds include prior SPK through **9812** lineage (**9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **9840** (+ prior **9812+9804+9792+9788+9768+9744**). Next constructive gap **2052** (OPEN).

Quantity	Value
9840 in SPK	**True** (Paley-I q=9839)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=9851->n=9852 Paley-I DIRECT q=9851 OR q=9859->n=9860 Paley-I DIRECT q=9859 OR q=9871->n=9872 Paley-I DIRECT q=9871 OR n=2052 standard SPK constructive gap – method-attack / lit OR Track B / Object 1 meter await